

Do Somatic Variant Callers Agree on the Same Tumor Sample?

A stratified analysis of multi-caller agreement by VAF, variant type and low-complexity context. Data:

1. Executive Summary

Question. When several somatic callers analyse the same tumor, how often do they all agree? Does agreement exceed 80% above 20% VAF, fall below 50% under 5% VAF, and is it lowest for indels and in low-complexity (LC) regions at every VAF?

Answer. The hypothesis is partly supported, and the verdict depends on what "variants" means.

VAF effect (H1/H2): strong and causal.

Low complexity (H4): clearly supported. Wherever the metric is not at its floor, LC regions (GIAB tandem repeats and homopolymers) disagreed more than other regions in every VAF bin. LC odds ratios for unanimity were:

No bin showed a significant reversal. Below 5-10% VAF, unanimity is 0 in both LC and non-LC regions, so "regardless of VAF" holds only in the trivial sense there.

Indels (H3): only conditionally supported.

"Indels are always worse regardless of VAF" is therefore not true as stated. The ranking depends on which callers are compared.

Practical implication. "All callers agree" is a high-precision filter for clonal variants. It silently discards almost every true subclonal (<10% VAF) variant, and much of the true signal in repeats. For low-VAF work, disagreement is mainly a sensitivity problem of the weakest caller (SomaticSniper, then MuSE and TNscope for SNVs; TNscope and Strelka for indels), so majority voting or caller-specific weighting is needed.

2. Research Question & Motivation

Hypothesis, decomposed and pre-registered in [planning.md](#)

Why it matters

Gap (from [literature_review.md](#))

Our contribution

3. Literature Summary (condensed)

- Chen 2020: Strelka2/Mutect2 SNV concordance >90% at 10% VAF, 75-89% at 5%, 20-41% at 1%.

Our numbers agree with these where they overlap (see §5.4).

4. Methodology

4.1 Data

| Experiment | Data | Callers compared | "Call" definition |

Exp 1 flag rules (PASS-like; see [code/somaticseq/somaticseq/annotate_caller.py](#)):

Exp 3 analysis rules:

Truth set and region annotations:

4.2 Denominators

4.3 Metrics and statistics

Metrics per stratum:

VAF bins: <5%, 5-10%, 10-20%, ≥20%.

Hypothesis tests:

Caller-set controls:

4.4 Environment and reproducibility

5. Results

5.1 H1/H2: agreement vs VAF (Figure 1, Figure 3)

![[fig1]](figures/fig1_agreement_by_vaf.png)

Table 1. All-callers fraction in SEQC2 (strict flags, native callers). Values are IL_T_1 [range over the 5 runs]. Full table: [results/d1_summary.csv](#).

| Denominator | Type | <5% | 5-10% | 10-20% | ≥20% |

Truth-set sizes per bin, IL_T_1: SNV 1,292 / 3,853 / 6,271 / 28,030; indel 102 / 287 / 422 / 814. Union sizes: SNV 83,570 / 74,665 / 37,199 / 156,082; indel 17,556 / 6,587 / 5,939 / 19,452.

H1 on the truth set. Supported (one-sided $p_{\text{Holm}} < 0.001$) for both types in the 4 standard-depth runs, with SNVs at 0.877±0.964 and indels at 0.865±0.909.

H1 on the union. Rejected in every run and in every configuration: raw PASS, HC-restricted and lenient. In IL_T_1, the ~20% SNV union contains only 18% truth-set sites, and 28% of its sites are single-caller calls.

H2. Supported on the truth set for SNVs in all runs (~0.3%). For indels it is supported in 4 of 5 runs (0.19±0.37); NS_380X is not significant (0.39, CI 0.30±0.49). It is supported everywhere on the union (~0.7%).

Majority agreement is much more forgiving (IL_T_1 truth, SNVs): 0.235 at <5%, 0.647 at 5±10%, 0.941 at 10±20%, 0.996 at ~20%.

Robustness (Exp 1b: raw VCFs, PASS only, TNscope added). The conclusions are unchanged:

The lenient flag definition also gives the same qualitative pattern (d1_summary.csv, flags=lenient).

Causal dilution test (Exp 2; Figure 3). Pooling truth variants over 6 purities, with the 4 shared callers, all-agree by expected VAF was:

Type	<5%	5±10%	10±20%	~20%
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95% CIs are within ±0.02 (d2_pooled_expected_vaf.csv).

For the same clonal variants (TVAF ~0.4), all-caller agreement (SNV: all 5; indel: all 4) fell with purity:

Purity	SNV (n=16,636)	Indel (n=330)
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For SNVs, 15,350 variants lost unanimity and only 1 gained it; McNemar $p < 1e-300$ (d2_paired_clonal.csv). Because the variants are identical, VAF alone drives the H1/H2 pattern.

![[fig3]](figures/fig3_purity_series.png)

TCGA MC3 (Exp 3; Figure 4).

The zeros below 10% are structural. MC3's SomaticSniper made no calls below 10% VAF (and only 0.30 at 10±20%), and VarScan2 made none below 5%. Every public <5% SNV is a MuTect+MuSE pair.

![[fig4]](figures/fig4_mc3.png)

5.2 H4: low-complexity regions (Figure 2)

![[fig2]](figures/fig2_lowcomplexity_by_vaf.png)

Table 2. All-callers fraction, IL_T_1, non-LC vs LC. Shared-3 callers (MuTect2, VarDict, Strelka) are used for both types.

Denominator	Type / region	<5%	5±10%	10±20%	~20%
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LC composition:

Within-bin tests. No bin in any dataset showed LC significantly better (results/verdicts_h3h4.csv).

Verdict counts across all contexts:

Experiment	Supported in all bins	Partial	Other
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The "partial" verdicts arise almost entirely because, below 5±10% VAF, all-caller agreement is ~0 in both LC and non-LC regions (a floor effect), or because truth LC cells are small.

Where the effect is measurable, it is large:

GLMs (results/d1_glm.csv), adjusted for depth, low mappability and segmental duplications. LC odds ratio for unanimity:

Denominator	SNV	Indel
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All ORs have $p < 1e-10$. The LC × VAF interaction was usually significant (LRT $p < 0.05$ in 28 of 30 native models). The LC penalty is therefore present at every measurable VAF, but its size changes with VAF rather than being constant. It is largest at 10±20% and ~20%.

5.3 H3: indels vs SNVs

Table 3. Within-bin difference in disagreement (indel ~ SNV), IL_T_1 unless noted. Brackets are 95% Newcombe CIs.

Context	<5%	5±10%	10±20%	~20%
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Notes on the table:

Findings:

![[fig5]](figures/fig5_caller_recall.png)

5.4 Comparison with the literature

6. Discussion

1. The denominator decides H1. Callers agree on real clonal mutations. They do not agree on what they call: 63±70% of all calls are single-caller calls, mostly low-VAF and LC artefacts (98% of union SNVs below 5% VAF are singletons). A statement like "callers agree on >80% of variants with VAF >20%" is true only for validated variants.

7. Limitations

- Truth-set bias. SEQC2 v1.2.1 was built from these same callers across replicates, and it excludes difficult regions. Truth-denominator agreement is therefore optimistic, and LC truth cells are small (e.g. 13 LC indels below 5% VAF). The union denominator avoids this bias but mixes true and false calls.

8. Validation

- Reproducibility. The D3 pipeline was re-run from scratch; d3_summary.csv and d3_cluster_bootstrap.csv were byte-identical (logs/d3_mc3_rerun.log). The other scripts are deterministic, since they involve no sampling.

9. Conclusions & Next Steps

Conclusions.

Recommendations.

Follow-ups.

References

- Alioto et al. 2015 Nat Commun.

Data sources:

Tools: bedtools 2.31, pysam, statsmodels, SomaticSeq (flag semantics).

Appendix: output files

- Per-site tables: